A simple adaptive control problem

Consider a system to be controlled

$$y = Gu, \qquad G = \frac{b}{s+a}$$

In time domain

$$(p+a)y(t) = bu(t), \qquad \dot{y}(t) = -ay(t) + bu(t)$$

We usually refer to the system to be controlled as the "Plant".

Control Objective

• Drive $y(t)$ to follow some reference command trajectory $y_m(t)$. For example, $y_m(t)$ may be generated by a reference system

$$\dot{y}_m(t) = -a_m y_m(t) + b_m r(t)$$

$a_m > 0$, i.e., stable reference model.

1. Design for a and b known

Consider a control input of the form

$$u(t) = k_1 y(t) + k_2 r(t)$$

Then the plant together with control (the closed-loop system) is

$$\begin{aligned} \dot{y}(t) &= -ay(t) + bu(t) \\ &= -ay(t) + b(k_1 y(t) + k_2 r(t)) \\ &= (-a + bk_1)y(t) + bk_2 r(t) \end{aligned}$$

The objective is $y \to y_m$ as $t \to \infty$, we compare with

$$\dot{y}_m(t) = -a_m y_m(t) + b_m r(t)$$

So the control parameters are chosen as

$$\begin{aligned} k_1 &= \frac{a - a_m}{b} \\ k_2 &= \frac{b_m}{b} \end{aligned}$$

Problem: We do not know a and b??

Adaptive Control: Find a way to achieve this automatically.

- Estimate a and b on-line, and try to use these estimates; (Will work most of the time); and

2. Design for a and b unknown

Since k_1 and k_2 are unknown, consider instead use the estimates to implement the adaptive controller.

$\hat{k}_1(t)$ is estimate of k_1
$\hat{k}_2(t)$ is estimate of k_2

Controller

$$u(t) = \hat{k}_1 y(t) + \hat{k}_2 r(t), \qquad Timevarying\ \ controller\ \ gains!!$$

Problem: How to choose $\{\hat{k}_1, \hat{k}_2$?}

$$
\begin{aligned}
\dot{y} &= -ay + bu \\
&= -ay + b(\hat{k}_1 y + \hat{k}_2 r) \\
&= -ay + b(k_1 - \tilde{k}_1)y + b(k_2 - \tilde{k}_2)r \\
&= (-a + bk_1)y - b\tilde{k}_1 y - b\tilde{k}_2 r + bk_2 r \\
&= -a_m y + b_m r - b\tilde{k}_1 y - b\tilde{k}_2 r
\end{aligned}
$$

where $-a + bk_1 = a_m$ and $bk_2 = b_m$.

$$
\tilde{k}_1 = k_1 - \hat{k}_1, \;\; \tilde{k}_2 = k_2 - \hat{k}_2,
$$

We have

$$
\begin{aligned}
e &= y - y_m \\
\dot{e} &= \dot{y} - \dot{y}_m = -a_m e - b\tilde{k}_1 y - b\tilde{k} r
\end{aligned}
$$

For this system, consider the Lyapunov function as

$$V(e, \tilde{k}_1, \tilde{k}) \;=\; \frac{1}{2}e^2 + \frac{|b|}{2\gamma_1}\tilde{k}_1^2 + \frac{|b|}{2\gamma_2}\tilde{k}^2$$

Taking the derivative of V

$$\begin{aligned}
\dot{V} &= e\dot{e} + \frac{|b|}{\gamma_1}\tilde{k}_1\dot{\tilde{k}}_1 + \frac{|b|}{\gamma_2}\tilde{k}_2\dot{\tilde{k}}_2 \\
&= -a_m e^2 - b\tilde{k}_1 ye - b\tilde{k}re - \frac{|b|}{\gamma_1}\tilde{k}_1\dot{\hat{k}}_1(t) - \frac{\dot{|b|}}{\gamma_2}\tilde{k}_2\dot{\hat{k}}_2(t) \\
&= -a_m e^2 - \frac{|b|}{\gamma_1}\tilde{k}_1[\dot{\hat{k}}_1(t) + sgn(b)\gamma_1 ye] - \frac{|b|}{\gamma_2}\tilde{k}_2[\dot{\hat{k}}_2(t) + sgn(b)\gamma_2 re]
\end{aligned}$$

To make $\dot{V} \leq 0$, Choose Adaptive Law: (Nonlinear updating mechanisms!!) Thus,

$$\boxed{\begin{array}{ll}
\dot{\hat{k}}_1 = -sgn(b)\gamma_1 e(t)y(t), & \gamma_1 > 0 \\
\dot{\hat{k}}_2 = -sgn(b)\gamma_2 e(t)r(t), & \gamma_2 > 0
\end{array}}$$

adaptive controllers contain both time-varying and nonlinear dynamics!!

$$\dot{V} \;=\; -a_m e^2 \leq 0$$

Collecting together the intermediate results:

- Results:

$$\dot{V} \;=\; -a_m e^2 \leq 0$$

- Quadratic form

$$V = \frac{1}{2}e^2 + \frac{|b|}{2\gamma_1}\tilde{k}_1^2 + \frac{|b|}{2\gamma_2}\tilde{k}^2$$

- This means that e is bounded, $\tilde{k}_1$ is bounded, $\tilde{k}$ is bounded.
  Note k_1 and k_2 are constants and

$$\begin{aligned}\tilde{k}_1 &\triangleq k_1 - \hat{k}_1 \\ \tilde{k}_2 &\triangleq k_2 - \hat{k}_2\end{aligned}$$

  Therefore $\hat{k}_1$ and $\hat{k}_2$ are bounded.

- Tracking error

$$e \triangleq y - y_m$$

$\because y_m(t)$ is a reference signal and obviously bounded.

$\therefore y(t)$ is bounded.

- $\because V$ positive definite.

  $\dot{V} = -a_m e^2 \leq 0$

  $\therefore V$ is bounded over all time.

$$\begin{aligned} \int_0^t \dot{V}(\tau)d\tau &= \int_0^t -a_m e^2(\tau)d\tau \\ \Rightarrow a_m \int_0^t e^2(\tau)d\tau &= V(0) - V(t) \end{aligned}$$

Since V is always bounded,

$\lim_{t\to\infty} \int_0^t e^2(\tau)d\tau \leq c_1$ a constant.

- $\dot{e} = -a_m e - b\tilde{k}_1 y - b\tilde{k} r$

  $\therefore \dot{e}(t)$ is bounded $\forall t \geq 0$.

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$$\left.\begin{array}{c} \dot{e} \quad bounded \\ \int_0^\infty e^2(\tau)d\tau \leq c_1 \end{array}\right\} \Rightarrow \lim_{t\to\infty} e(t) = 0$$

$$\lim_{t\to\infty} y(t) - y_m(t) = 0$$

The adaptive controller achieves the designed objective!

Rigorous analysis of adaptive control is typically as difficult.

Summary

Plant:

$$\dot{y}(t) = -ay(t) + bu(t)$$

Reference Model:

$$\dot{y}_m(t) = -a_m y_m(t) + b_m r(t), \;\; a_m > 0$$

Control Law:

$$u(t) = \hat{k}_1 y(t) + \hat{k}_2 r(t), \qquad Timevarying$$

Time varying Adaptive Law:

$$\begin{aligned} \dot{\hat{k}}_1 &= -sgn(b)\gamma_1 e(t)y(t), \qquad \gamma_1 > 0 \\ \dot{\hat{k}}_2 &= -sgn(b)\gamma_2 e(t)r(t), \qquad \gamma_2 > 0 \\ e(t) &= y(t) - y_m(t) \end{aligned}$$

Result:

If the adaptive controller is applied to the plant, then all signals $\{y, u, \hat{k}_1, \hat{k}_2\}$ **are bounded and** $\lim_{t\to\infty}[y(t) - y_m(t)] = 0$**.**